

On the Complexity of Some Variations of Y -Dominating Functions on Graphs

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Abstract

Let Y be a subset of real numbers. A Y -dominating function of a graph $G = (V, E)$ is a function $f : V \rightarrow Y$ such that $\sum_{u \in N_G[v]} f(u) \geq 1$ for all vertices $v \in V$, where $N_G[v] = \{v\} \cup \{u \mid (u, v) \in E\}$. Let $f(S) = \sum_{u \in S} f(u)$ for any subset S of V and let $f(V)$ be the weight of f . The Y -domination problem is to find a Y -dominating function of minimum weight for a graph. In this paper, we study the variations of Y -domination such as $\{k\}$ -domination, k -tuple domination, signed domination, and minus domination for some classes of graphs. We present a unified approach to these four domination problems on strongly chordal graphs. Notice that trees, block graphs, interval graphs, and directed path graphs are subclasses of strongly chordal graphs. This paper also gives complexity results for these four domination problems on doubly chordal graphs, dually chordal graphs, chordal bipartite graphs, and planar graphs.

1 Introduction

Let $G = (V, E)$ be a finite, undirected, simple graph. For any vertex $v \in V$, the *open neighborhood* of v in G is $N_G(v) = \{u \in V \mid (u, v) \in E\}$ and the *closed neighborhood* of v in G is $N_G[v] = N_G(v) \cup \{v\}$. The *degree* of a vertex v in G is $\deg_G(v) = |N_G(v)|$. We use $\delta(G)$ to denote the minimum degree of a vertex in G . We use $V(G)$ and $E(G)$ to denote the vertex set and the edge set of G , respectively. A *dominating set* of a graph $G = (V, E)$ is a subset D of V such that $|N_G[v] \cap D| \geq 1$ for all vertices $v \in V$. The *domination number* $\gamma(G)$ is the minimum cardinality of a dominating set of G . The *domination problem* is to find a dominating set of minimum cardinality

for a graph G . Let Y be a subset of real numbers. Let $f : V \rightarrow Y$ be a function which assigns to each $v \in V$ a value in Y . The set Y is called the *weight set* of f . We let $f(S) = \sum_{u \in S} f(u)$ for any subset S of V and let $f(V)$ be the *weight* of f . The function f of G is called a *Y -dominating function* if $f(N_G[v]) \geq 1$ for every vertex $v \in V$. The Y -domination problem is to find a Y -dominating function of minimum weight for a graph. The Y -domination problem was introduced by Bange et al. as a generalization of domination [1]. We can view a dominating set as a Y -dominating function f with $Y = \{0, 1\}$. In this case, the function f is called a *dominating function* and $\gamma(G) = \min\{f(V) \mid f \text{ is a dominating function of } G\}$.

Domination and its variations have been extensively studied in the literature and surveyed in [9, 21, 22]. Among the variations of domination, there is an extensive number of papers concerning Y -domination and its variations. In this paper, we restrict our attention to the variations of Y -domination such as k -tuple domination, $\{k\}$ -domination, signed domination, and minus domination on graphs.

Definition 1. For a fixed positive integer k , a *k -tuple dominating set* of a graph $G = (V, E)$ is a subset D of V such that $|N_G[v] \cap D| \geq k$ for all vertices $v \in V$. The *k -tuple domination number* $\gamma_{\times k}(G)$ is the minimum cardinality of a k -tuple dominating set of G .

Remark 1. A graph G has a k -tuple dominating set if and only if $\delta(G) + 1 \geq k$. A k -tuple dominating set of a graph $G = (V, E)$ can be viewed as a *k -tuple dominating function* $f : V \rightarrow \{0, 1\}$ such that $f(N_G[v]) \geq k$ for all vertices $v \in V$. Then, the k -tuple domination number of G is equal to the minimum weight of a k -tuple dominating function of G .

Definition 2. A *signed dominating function* of a

graph $G = (V, E)$ is a function $f : V \rightarrow \{-1, 1\}$ such that $f(N_G[v]) \geq 1$ for all vertices $v \in V$. The *signed domination number* of G is $\gamma_s(G) = \min\{f(V) \mid f \text{ is a signed dominating function of } G\}$.

Definition 3. A *minus dominating function* of a graph $G = (V, E)$ is a function $f : V \rightarrow \{-1, 0, 1\}$ such that $f(N_G[v]) \geq 1$ for all vertices $v \in V$. The *minus domination number* of G is $\gamma^-(G) = \min\{f(V) \mid f \text{ is a minus dominating function of } G\}$.

Definition 4. For a fixed positive integer k , a *$\{k\}$ -dominating function* of a graph $G = (V, E)$ is a function $f : V \rightarrow \{0, 1, 2, \dots, k\}$ such that $f(N_G[v]) \geq k$ for all vertices $v \in V$. The *$\{k\}$ -domination number* of G is $\gamma_{\{k\}}(G) = \min\{f(V) \mid f \text{ is a } \{k\}\text{-dominating function of } G\}$.

The k -tuple domination problem is to find a k -tuple dominating set of a graph G of minimum cardinality. The $\{k\}$ -domination (respectively, signed domination and minus domination) problem is to find a $\{k\}$ -dominating (respectively, signed dominating and minus dominating) function of a graph G of minimum weight. By definition, domination is a special case of k -tuple domination and $\{k\}$ -domination when $k = 1$ and thus $\gamma(G) = \gamma_{\times 1}(G) = \gamma_{\{1\}}(G)$.

In this paper, we present a unified approach to these four problems on strongly chordal graphs. This paper also gives complexity results for (1) the $\{k\}$ -domination problem on dually chordal graphs, (2) the k -tuple and signed domination problems on doubly chordal graphs, and (3) the k -tuple domination, signed domination, and minus domination problems on chordal bipartite graphs and planar graphs. These problems have been open for at least five years. This paper presents the following results. In Section 2, we show that the $\{k\}$ -domination problem on dually chordal graphs is linear-time solvable. In Section 3, we introduce the concept of the *labelled domination problem*. The labelled domination problem includes the $\{k\}$ -domination, k -tuple domination, minus domination and signed domination problems as special cases. Although the concept of labelled domination leads to a difficult problem, however, we shall show that the problem can be solved in linear-time for strongly chordal graphs. Notice that trees, block graphs, interval graphs, and directed path graphs are subclasses of strongly chordal graphs [5]. In Section 4, we discuss the relationship between the *generalized clique-transversal problem* [8] and the labelled domination problem on strongly chordal graphs. In Section 5, we present several NP-complete results for

the k -tuple and signed domination problems on doubly chordal graphs, and the k -tuple domination, signed domination, and minus domination problems on chordal bipartite graphs and planar graphs. It is worth mentioning that the $\{k\}$ -domination and k -tuple domination problems with $k > 1$ reflect distinct time complexities on dually chordal graphs (respectively, doubly chordal graphs). For a fixed positive integer $k > 1$, the $\{k\}$ -domination problem is linear-time solvable for doubly chordal graphs and dually chordal graphs but the k -tuple domination problem is NP-complete for these two classes of graphs. However, the $\{k\}$ -domination and k -tuple domination problems are linear-time solvable for these two classes of graphs when $k = 1$.

Section 6 gives the concluding remarks in the last section. Definitions of graph classes and their relevant properties will be given in the corresponding sections. For detailed terminology, notations, and definitions and properties of graphs not given in our paper we refer to [5].

2 The $\{k\}$ -domination problem on dually chordal graphs

Let G be a graph and K be a positive integer. For a fixed positive integer $k > 1$, the problem of determining whether $\gamma_{\{k\}}(G) \leq K$ is NP-complete [16]. In this section we show that the $\{k\}$ -domination problem is linear-time solvable on dually chordal graphs. We recall the definitions and some results on dually chordal graphs.

Given a graph $G = (V, E)$ and a vertex $v \in V$, a vertex $u \in N_G[v]$ is a *maximum neighbor* of v if for all $w \in N_G[v]$, $N_G[w] \subseteq N_G[u]$ holds (note that $u = v$ is not excluded). Let G_i be the subgraph of G induced by $\{v_i, v_{i+1}, \dots, v_n\}$ and let $N_i[v]$ denote the closed neighborhood of v in G_i . A linear ordering $(v_1, v_2, \dots, v_n)$ of V is a *maximum neighborhood ordering* of G if for all $i \in \{1, \dots, n\}$, there is a maximum neighbor $u_i \in N_i[v_i]$; i.e.,

$$\text{for all } w \in N_i[v_i], N_i[w] \subseteq N_i[u_i] \text{ holds.}$$

A graph G is *dually chordal* [4] if and only if G has a maximum neighborhood ordering. A graph G is *doubly chordal* if and only if G is chordal and dually chordal [27]. Maximum neighborhood orderings can be computed in linear-time [13]. For dually chordal graphs, the domination problem can be solved in linear-time [3].

Definition 5. The *distance* $d_G(u, v)$ between vertices u, v of a graph $G = (V, E)$ is the number of edges of a shortest path from u to v . A dominating set of G is equivalent to a subset $D \subseteq V$ such that for any vertex $v \in V$ there is a vertex $u \in D$ with $d_G(u, v) \leq 1$. A *2-packing set* is a subset $P \subseteq V$ such that $d_G(u, v) > 2$ for every pair of vertices $u, v \in P$. The maximum cardinality of a 2-packing set of G is called the *2-packing number* and denoted by $P_2(G)$. It is clear that $P_2(G) \leq \gamma(G)$ for any graph G .

The following theorem is deduced from the result of Brandstädt et al. [3].

Theorem 1. *For any dually chordal graph G , $\gamma(G) = P_2(G)$.*

Theorem 2. [6] *Given a graph $G = (V, E)$ and an integer $k \geq 1$, $\gamma_{\{k\}}(G) = k\gamma(G)$ if $P_2(G) = \gamma(G)$.*

Proof. Suppose that D is a dominating set of a graph $G = (V, E)$ of $\gamma(G)$ vertices. Let $f : V \rightarrow \{0, 1, 2, \dots, k\}$ be a function defined by $f(v) = k$ if $v \in D$ and $f(v) = 0$ if $v \in V \setminus D$. Notice that $|N_G[v] \cap D| \geq 1$ for every vertex $v \in V$. Therefore, $f(N_G[v]) \geq k$ for every vertex $v \in V$ and thus f is a $\{k\}$ -dominating function of G . We have $\gamma_{\{k\}}(G) \leq k\gamma(G)$.

Suppose that P is a 2-packing set of G of $P_2(G)$ vertices and $P_2(G) = \gamma(G)$. For every pair of vertices $u, v \in P$, $d_G(u, v) > 2$. The distance between u and v is at least 3. Let $f : V \rightarrow \{0, 1, \dots, k\}$ be a minimum $\{k\}$ -dominating function of G . For every vertex $v \in P$, $f(N_G[v]) \geq k$. Since $N_G[u] \cap N_G[v] = \emptyset$ for $u, v \in P$, therefore $\gamma_{\{k\}}(G) \geq k \cdot P_2(G) = k\gamma(G)$. Following the discussion above, $\gamma_{\{k\}}(G) = k\gamma(G)$ if $P_2(G) = \gamma(G)$. $\square$

By Theorems 1–2, we can obtain a minimum dominating set D of a dually chordal graph $G = (V, E)$ by the algorithm of [3], and then a minimum $\{k\}$ -dominating function f of G can be obtained by setting $f(v) = k$ if $v \in D$ and $f(v) = 0$ if $v \in V \setminus D$. We have the following theorem.

Theorem 3. *The $\{k\}$ -domination problem on dually chordal graphs can be solved in linear-time.*

3 A unified approach to some variations of domination on strongly chordal graphs

Let G be a graph and K be a positive integer. The problem of determining whether $\gamma(G) \leq K$ (respectively, $\gamma^-(G) \leq K$ and $\gamma_s(G) \leq K$) is NP-complete even when restricted to bipartite graphs and chordal graphs [2, 11, 20]. For a fixed positive integer $k > 1$, the problem of determining whether $\gamma_{\times k}(G) \leq K$ is NP-complete even when restricted to bipartite graphs and split graphs [26]. For trees, the minus, signed, and k -tuple domination problems are linear-time solvable [11, 20, 25]. For cographs and k -trees with bounded k , the signed domination problem can be solved in polynomial-time by using the algorithms of [32]. Besides, the algorithms of [32] can be easily modified to solve the minus domination problem in polynomial-time for cographs and k -trees with bounded k . In [17], Gong presented linear-time algorithms for finding the signed domination numbers of some special block graphs. For strongly chordal graphs, the k -tuple domination problem can be solved in linear-time [10, 26]. In [10], Chen proposed a generalization of k -tuple domination called *fault tolerant domination*.

Definition 6. Let $G = (V, E)$ be a graph. Let $h : V \rightarrow \mathbb{N}$ be a function which assigns to each $v \in V$ a positive integer. A *fault tolerant dominating set* is a subset D of V such that $|N_G[v] \cap D| \geq h(v)$ for all vertices $v \in V$. The fault tolerant domination problem is to find a fault tolerant dominating set of minimum cardinality for a graph.

Chen solved the fault tolerant domination problem in linear-time for strongly chordal graphs and showed the following result.

Theorem 4. *Given a function $g : V \rightarrow \{-1, 1\}$ of a graph $G = (V, E)$, let $S = \{v | g(v) = 1, v \in V\}$. The function g is a signed dominating function if and only if $|S \cap N_G[v]| \geq \lfloor \frac{|N_G[v]|}{2} \rfloor + 1$ for all vertices $v \in V$.*

By Theorem 4, Chen solved the signed domination problem on a strongly chordal graph $G = (V, E)$ in linear-time using the algorithm for the fault tolerant domination problem on G with $h(v) = \lfloor \frac{|N_G[v]|}{2} \rfloor + 1$ for all vertices $v \in V$.

In this section, we develop a unified approach to the $\{k\}$ -domination, k -tuple domination, signed domination and minus domination problems on strongly chordal graphs. We recall the definitions and some properties of strongly chordal graphs.

Let $G = (V, E)$ be a graph. A *clique* is a subset of pairwise adjacent vertices of V . A vertex v is *simplicial* if all vertices of $N_G[v]$ form a clique. The ordering $(v_1, v_2, \dots, v_n)$ of the vertices of V is a *perfect elimination ordering* of G if for all $i \in \{1, \dots, n\}$, v_i is a simplicial vertex of the subgraph G_i of G induced by $\{v_i, v_{i+1}, \dots, v_n\}$. Let $N_i[v]$ denote the closed neighborhood of v in G_i . Rose [30] showed the characterization that a graph is *chordal* if and only if it has a perfect elimination ordering. A perfect elimination ordering is called a *strong elimination ordering* if it has the following property:

For $i \leq j \leq k$ if v_j and v_k belong to $N_i[v_i]$ in G_i , then $N_i[v_j] \subseteq N_i[v_k]$.

Farber [14] showed that a graph is *strongly chordal* if and only if it admits a strong elimination ordering. So far, the fastest algorithm to recognize a strongly chordal graph and give a strong elimination ordering takes $O(m \log n)$ [29] or $O(n^2)$ time [31]. Strongly chordal graphs form a subclass of doubly chordal graphs [4] and a superclass of trees, block graphs, interval graphs, and directed path graphs.

To develop a unified approach to the $\{k\}$ -domination, k -tuple domination, signed domination and minus domination problems on strongly chordal graphs, we introduce the concept of *labelled domination*.

Definition 7. Let ℓ, d, I_1 be integers and $\ell, d > 0$. Let Y be the set $\{I_1, I_1 + d, I_1 + 2d, \dots, I_1 + (\ell - 1) \cdot d\}$. Suppose that $G = (V, E)$ is a graph and L is a labelling function which assigns to each $v \in V$ a label $L(v) = (t(v), k(v))$, where $t(v) \in Y \cup \{F\}$ and $k(v)$ is an integer. An *L -dominating function* of $G = (V, E)$ is a function $f : V \rightarrow Y$ satisfying the following two conditions.

- (1) If $t(v) \neq F$, then $f(v) = t(v)$.
- (2) $f(N_G[v]) \geq k(v)$ for all vertices $v \in V$.

The *L -domination number* $\gamma_L(G)$ is the minimum weight of an L -dominating function of G . The *labelled domination problem* is to find an L -dominating function of G of minimum weight. By definition of the L -dominating function of a graph $G = (V, E)$, the following statements are straightforward.

- (1) If $Y = \{0, 1\}$ and every vertex $v \in V$ is associated with a label $L(v) = (F, k)$, then a minimum L -dominating function $f : V \rightarrow Y$ of G is equivalent to a minimum k -tuple dominating function of G .

- (2) If $Y = \{0, 1, 2, \dots, k\}$ and every vertex $v \in V$ is associated with a label $L(v) = (F, k)$, then a minimum L -dominating function $f : V \rightarrow Y$ of G is equivalent to a minimum $\{k\}$ -dominating function of G .
- (3) If $Y = \{-1, 1\}$ and every vertex $v \in V$ is associated with a label $L(v) = (F, 1)$, then a minimum L -dominating function $f : V \rightarrow Y$ of G is equivalent to a minimum signed dominating function of G .
- (4) If $Y = \{-1, 0, 1\}$ and every vertex $v \in V$ is associated with a label $L(v) = (F, 1)$, then a minimum L -dominating function $f : V \rightarrow Y$ of G is equivalent to a minimum minus dominating function of G .

By the statements above, the labelled domination problem includes the $\{k\}$ -domination, k -tuple domination, minus domination and signed domination problems as special cases. Although the concept of labelled domination leads to a difficult problem, however, we shall show that the problem can be solved efficiently on strongly chordal graphs. In the following, we give Algorithm $\text{MLD}(G, L, I_1, \ell, d)$ to solve the labelled domination problem in linear-time for a strongly chordal graph G if the strong elimination ordering is given.

Algorithm $\text{MLD}(G, L, I_1, \ell, d)$ takes G, L, I_1, ℓ , and d as inputs. Input G represents a strongly chordal graph $G = (V, E)$ with a strong elimination ordering $(v_1, v_2, \dots, v_n)$. Inputs ℓ, d, I_1 are integers and $\ell, d > 0$. The weight set Y is assumed to be the set $\{I_1, I_1 + d, I_1 + 2d, \dots, I_1 + (\ell - 1) \cdot d\}$. Input L represents a labelling function which assigns to every vertex v of G a label $L(v) = (t(v), k(v))$, where $t(v) \in Y \cup \{F\}$ and $k(v)$ is an integer. The algorithm finds a minimum L -dominating function of G .

Algorithm $\text{MLD}(G, L, I_1, \ell, d)$ starts with initializing a function f such that for each vertex $v \in V$,

$$f(v) = \begin{cases} I_1 + (\ell - 1) \cdot d & \text{if } t(v) = F, \\ t(v) & \text{otherwise.} \end{cases}$$

The algorithm determines whether G has an L -dominating function by checking if the function f is an L -dominating function of G . If there is a vertex v with $k(v) > f(N_G[v])$, then f is not an L -dominating function of G , and the algorithm stops and returns the infeasibility of the problem. Otherwise, f is an L -dominating function of G and the algorithm processes vertices in strong elimination ordering $v_1, v_2, \dots, v_n$ to reduce the

weight of the function f . To do so, if a vertex v_i is with $t(v_i) = F$, the algorithm computes the value $M = \min\{f(N_G[v]) - k(v) \mid v \in N_G[v_i]\}$ and then assigns the value $\max\{I_1, I_1 + (\lceil \ell - \frac{M}{d} \rceil - 1) \cdot d\}$ to the vertex v_i . We summarize the procedure described above in the following algorithm:

Algorithm MLD(G, L, I_1, ℓ, d)

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/* Initialization */
1. for  $i = 1$  to  $n$  do
2.   if  $t(v_i) = F$  then  $f(v_i) = I_1 + (\ell - 1) \cdot d$ .
3.   else  $f(v_i) = t(v_i)$ .
4. endfor.
/* Determining whether  $G$  has an  $L$ -dominating
   function. */
5. for  $i = 1$  to  $n$  do
6.   if  $k(v_i) > f(N_G[v_i])$  then stop and return
      the infeasibility of the problem.
7. endfor.
/* Reducing the weight of the function */
8. for  $i = 1$  to  $n$  do
9.   if  $t(v_i) = F$ , then
10.     $M = \min\{f(N_G[v]) - k(v) \mid v \in N_G[v_i]\}$ .
11.     $f(v_i) = \max\{I_1, I_1 + (\lceil \ell - \frac{M}{d} \rceil - 1) \cdot d\}$ .
12. endfor.
13. return the function  $f$ .

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Lemma 1. *If the function f initialized by Algorithm MLD in Steps 1–4 is not an L -dominating function of a strongly chordal graph $G = (V, E)$, then G has no L -dominating function.*

Proof. For $1 \leq i \leq n$, the function f initialized by Algorithm MLD in Steps 1–4 assigns the maximum value, $I_1 + (\ell - 1) \cdot d$, in Y to vertex v_i if $t(v_i) = F$ and assigns $t(v_i)$ to vertex v_i if $t(v_i) \neq F$. By the definition of L -dominating functions, every L -dominating function of G assigns $t(v_i)$ to vertex v_i if $t(v_i) \neq F$. Therefore, f has the maximum weight among all L -dominating functions of G if f is an L -dominating function of G . As a result, if there is a vertex $v_i \in V$ with $k(v_i) > f(N_G[v_i])$, then f is not an L -dominating function of G and we can not find any L -dominating function in G . $\square$

Lemma 2. *The function f returned from Step 13 of Algorithm MLD is an L -dominating function of a strongly chordal graph G .*

Proof. By Algorithm MLD, the function f initialized in Steps 1–4 is an L -dominating function of G if the algorithm does not stop in Step 6. In Steps 8–12, the algorithm processes vertices in strong elimination ordering $v_1, v_2, \dots, v_n$ to reduce the weight of the function f . Obviously, the function f at the beginning of the first iteration of Steps 8–12 is an L -dominating function of G . We assume that the function f of i -th iteration of Steps 8–12 is an L -dominating function of G for $1 \leq i \leq n$. In the following, we show that at the end of the i -th iteration of Steps 8–12, the new function f obtained by changing the value of $f(v_i)$ in Step 11 is still an L -dominating function of G . Notice that the function f at the end of i -th iteration of Steps 8–12 is the function f at the beginning of the $(i + 1)$ -th iteration.

Suppose that $t(v_i) \neq F$. It is clear that the function f at the end of the i -th iteration is the same as the function f at the beginning of the i -th iteration. Suppose that $t(v_i) = F$. Let $x = \max\{I_1, I_1 + (\lceil \ell - \frac{M}{d} \rceil - 1) \cdot d\}$. Then $x \geq I_1 + (\lceil \ell - \frac{M}{d} \rceil - 1) \cdot d$. We have

$$x \geq I_1 + (\ell - 1 - \frac{M}{d}) \cdot d \Rightarrow M \geq (I_1 + (\ell - 1) \cdot d) - x.$$

Since $M = \min\{f(N_G[v]) - k(v) \mid v \in N_G[v_i]\}$, therefore $f(N_G[v]) - k(v) \geq M \geq (I_1 + (\ell - 1) \cdot d) - x$ for every vertex $v \in N_G[v_i]$. We have $f(N_G[v]) - (I_1 + (\ell - 1) \cdot d) + x \geq k(v)$ for every vertex $v \in N_G[v_i]$. Notice that $f(v_i) = I_1 + (\ell - 1) \cdot d$ before the execution of Step 11. Therefore, the new function f obtained by replacing the value of $f(v_i)$ with x in Step 11 is still an L -dominating function of G . Following the discussion above, we know that the function f returned from Step 13 of Algorithm MLD is an L -dominating function of a strongly chordal graph G . $\square$

Lemma 3. *Let ℓ, d, I_1 be integers and $\ell, d > 0$. Let Y be the set $\{I_1, I_1 + d, I_1 + 2d, \dots, I_1 + (\ell - 1) \cdot d\}$. The function $f : V \rightarrow Y$ returned from Algorithm MLD(G, L, I_1, ℓ, d) is a minimum L -dominating function of a strongly chordal graph $G = (V, E)$.*

Proof. By Lemma 2, the function f returned from Algorithm MLD is an L -dominating function of G . In the following, we show that f is a minimum L -dominating function of G . Among all minimum L -dominating functions of G , we let h be

a minimum L -dominating function of G such that the number of vertices in $\{v|v \in V, f(v) = h(v)\}$ is maximum. We claim that $f(v) = h(v)$ for every vertex $v \in V$. Assume for contrary that W is a nonempty set of all vertices w with $f(w) \neq h(w)$. Suppose that t is the smallest index such that $v_t \in W$. Obviously, $t(v_t) = F$. We consider the following cases.

Case 1: $h(v_t) < f(v_t)$. The value of $f(v_t)$ is finalized in Step 11 at t -iteration of Algorithm MLD (where an iteration here of the algorithm is understood as one iteration of Steps 8–12). At t -th iteration of Algorithm MLD, $M = \min\{f(N_G[v]) - k(v) | v \in N_G[v_t]\}$ and $f(v_t) = \max\{I_1, I_1 + (\lceil \ell - \frac{M}{d} \rceil - 1) \cdot d\}$. We consider the following two cases:

Case 1.1: $f(v_t) = I_1$. Then $h(v_t) < f(v_t) = I_1$, which contradicts the assumption that $h(v_t) \in Y$ since I_1 is the smallest number in Y .

Case 1.2: $f(v_t) = I_1 + (\lceil \ell - \frac{M}{d} \rceil - 1) \cdot d$. Then $h(v_t) \leq f(v_t) - d = I_1 + (\lceil \ell - \frac{M}{d} \rceil - 2) \cdot d$. Let v_c be a vertex in $N_G[v_t]$ such that $M = f(N_G[v_c]) - k(v_c)$. Notice that $f(v_t) = I_1 + (\ell - 1) \cdot d$ before the execution of Step 11 at the t -th iteration. Therefore, $f(N_G[v_c] \setminus \{v_t\}) = M + k(v_c) - (I_1 + (\ell - 1) \cdot d)$ before the execution of Step 11 at the t -iteration. Since only the value of $f(v_t)$ was changed at the t -th iteration, therefore $f(N_G[v_c] \setminus \{v_t\})$ is still equal to $M + k(v_c) - (I_1 + (\ell - 1) \cdot d)$ at the end of t -th iteration.

Notice that $h(v_x) = f(v_x)$ for every index $x < t$. At the end of the t -th iteration, $h(v_x) \leq f(v_x) = I_1 + (\ell - 1) \cdot d$ for every index $x > t$. Then,

$$\begin{aligned} h(N_G[v_c]) &\leq f(N_G[v_c] \setminus \{v_t\}) + h(v_t) \\ &\leq f(N_G[v_c] \setminus \{v_t\}) + I_1 \\ &\quad + (\lceil \ell - \frac{M}{d} \rceil - 2) \cdot d \\ &= M + k(v_c) - (I_1 + (\ell - 1) \cdot d) \\ &\quad + (I_1 + (\lceil \ell - \frac{M}{d} \rceil - 2) \cdot d) \\ &< M + k(v_c) - (I_1 + (\ell - 1) \cdot d) \\ &\quad + (I_1 + ((\ell - \frac{M}{d}) - 1) \cdot d) \\ &= M + k(v_c) - (I_1 + (\ell - 1) \cdot d) \\ &\quad + (I_1 + (\ell - 1) \cdot d - M) \\ &= k(v_c) \end{aligned}$$

Hence $h(N_G[v_c]) < k(v_c)$ which contradicts the assumption that h is an L -dominating function of G .

Case 2: $f(v_t) < h(v_t)$. Let $Y = \{a_1, a_2, \dots, a_\ell\}$ where $a_1 = I_1$, $a_2 = I_1 + d$, $a_3 = I_2 + 2d, \dots, a_\ell = I_1 + (\ell - 1) \cdot d$. We let $f(v_t) = a_i$ and $h(v_t) = a_j$ for $1 \leq i < j \leq \ell$. Let $P = \{v|v \in N_G[v_t], h(N_G[v]) - a_j + a_i < k(v)\}$. We have that $P \neq \emptyset$. Otherwise, $h(N_G[v]) - a_j + a_i \geq k(v)$ for every $v \in N_G[v_t]$ and there is an L -dominating function g with

$g(V) < h(V)$ by setting $g(v_t) = h(v_t) - a_j + a_i = a_i$ and $g(v) = h(v)$ for every vertex $v \in V \setminus \{v_t\}$, a contradiction to the assumption that h is a minimum L -dominating function of G .

Notice that $h(v_x) = f(v_x)$ for every index $x < t$. Since $h(N_G[v]) - a_j + a_i < k(v)$ and $f(N_G[v]) \geq k(v)$ for every vertex $v \in P$, therefore $N_G[v] \cap \{v_x|v_x \in W, t < x, \text{ and } h(v_x) < f(v_x)\} \neq \emptyset$ for every vertex $v \in P$.

Let s be the smallest index of vertices in P . Let b be the smallest index of $N_G[v_s] \cap \{v_x|v_x \in W, t < x, \text{ and } h(v_x) < f(v_x)\}$. Notice that $P \subseteq N_G[v_t]$. Since $h(v_t), f(v_t), h(v_b)$, and $f(v_b)$ are in Y , there exist two positive integers c_1 and c_2 such that $h(v_t) = f(v_t) + c_1 \cdot d$ and $f(v_b) = h(v_b) + c_2 \cdot d$. We define a function h' as follows.

- (1) If $c_1 \leq c_2$, $h'(v_t) = h(v_t) - c_1 \cdot d = f(v_t)$, $h'(v_b) = h(v_b) + c_1 \cdot d$ and $h'(v) = h(v)$ for every vertex $v \in V \setminus \{v_t, v_b\}$.
- (2) If $c_1 > c_2$, $h'(v_t) = h(v_t) - c_2 \cdot d$, $h'(v_b) = h(v_b) + c_2 \cdot d = f(v_b)$, and $h'(v) = h(v)$ for every vertex $v \in V \setminus \{v_t, v_b\}$.

Clearly, $h(V) = h'(V)$ and $|\{v|v \in V, f(v) = h'(v)\}| \geq |\{v|v \in V, f(v) = h(v)\}| + 1$. We prove $h'(N_G[v]) \geq k(v)$ for every vertex $v \in V$ by showing that $P \subseteq N_G[v_b]$. There are two cases:

Case 2.1: $s \leq t$. Then $s \leq t < b$. By definition of the strong elimination ordering, $N_s[v_t] \subseteq N_s[v_b]$. Since $P \subseteq N_s[v_t]$, we have $P \subseteq N_G[v_b]$.

Case 2.2: $s > t$. By definition of the strong elimination ordering, $N_t[v_s] \subseteq N_t[v]$ for every vertex $v \in P$. Since $v_b \in N_t[v_s]$, $v_b \in N_t[v]$ for every vertex $v \in P$. In other words, $P \subseteq N_G[v_b]$.

Hence, h' is a minimum L -dominating function such that the number of vertices in $\{v|v \in V, f(v) = h'(v)\}$ is larger than that of vertices in $\{v|v \in V, f(v) = h(v)\}$, a contradiction to the assumption that the number of vertices in $\{v|v \in V, f(v) = h(v)\}$ is maximum.

Following the discussion above, W does not exist. Hence, f is a minimum L -dominating function of G . $\square$

Theorem 5. Let $G = (V, E)$ be a strongly chordal graph with $|V| = n$ and $|E| = m$. Algorithm MLD solves the labelled domination problem on G in $O(n + m)$ time if the strong elimination ordering is given.

Proof. In a practical implementation of Algorithm MLD, we use $d(v_i)$ to keep track of $f(N_G[v_i])$ for each vertex $v_i \in V$ and use $m(v_i)$ to keep track of $d(v_i) - k(v_i)$. Let $F(v_i)$ denote

the set $\{v | v \in N_G[v_i], t(v) = F\}$ for every vertex $v_i \in V$. Following the initialization of a function f in Steps 1–4, we initialize

$$d(v_i) = \left(\sum_{j=1}^{|F(v_i)|} (I_1 + (\ell - 1) \cdot d) + \sum_{\substack{v \in N_G[v_i] \\ t(v) \neq F}} t(v) \right)$$

and

$$m(v_i) = d(v_i) - k(v_i).$$

The initialization of $d(v_i)$ and $m(v_i)$ can be done in $O(\deg_G(v_i) + 1)$ time. While $f(v_i)$ is replaced by a number $x = \max\{I_1, I_1 + (\lceil \ell - \frac{M}{d} \rceil - 1) \cdot d\}$, $d(v)$ and $m(v)$ are respectively decreased by $(I_1 + (\ell - 1) \cdot d) - x$ for every vertex $v \in N_G[v_i]$. This can be done in $O(\deg_G(v_i) + 1)$ time. At i -th iteration, $1 \leq i \leq n$, M can be computed in $O(\deg_G(v_i) + 1)$ time by verifying $m(v)$ for every vertex $v \in N_G[v_i]$. Hence, the running time of Algorithm MLD is $O(\sum_{v_i \in V} (\deg(v_i) + 1)) = O(n + m)$. $\square$

3.1 Relationship between the generalized clique-transversal and labelled domination problems on strongly chordal graphs

In this section we discuss the relationship between the generalized clique-transversal and labelled domination problems. We show that Algorithm MLD can be used to solve the generalized clique-transversal problem in linear-time for strongly chordal graphs.

Definition 8. Let $G = (V, E)$ be a graph. A clique is a subset of pairwise adjacent vertices of G . A *maximal clique* is a clique that is not a proper subset of any other clique. Let $C(G)$ denote the set of all maximal cliques of G . Let $R = \{(Q_i, r_i) | Q_i \in C(G)\}$, in which each element (Q_i, r_i) indicates that the maximal clique Q_i is associated with an integer r_i and $0 \leq r_i \leq |Q_i|$. An *R -clique-transversal set* of G is a subset D of V such that $|D \cap Q_i| \geq r_i$ for every $Q_i \in C(G)$. The minimum cardinality $\tau_R(G)$ of an R -clique-transversal set of G is called the *R -clique-transversal number* of G . The *generalized clique-transversal problem* is to find an R -clique-transversal set of G of minimum cardinality.

In [8], Chang et al. solved the generalized clique-transversal problem on strongly chordal graphs in linear-time if strongly elimination orderings are given. In the following, we show a

linear-time reduction from the generalized clique transversal problem on strongly chordal graphs to the labelled domination problem on strongly chordal graphs.

Let $G = (V, E)$ be a strongly chordal graph and $C(G) = \{Q_1, Q_2, \dots, Q_p\}$. In [18], Guruswami et al. construct a graph called the *vertex-clique incidence graph* H of G as follows:

The vertex set of H is $V \cup X$ where $X = \{x_1, x_2, \dots, x_p\}$. In H , (1) X is an independent set, (2) two vertices of V are adjacent if they are adjacent in G , and (3) for $1 \leq i \leq p$, $x_i \in X$ is adjacent to $v_j \in V$ if $v_j \in Q_i$ in G .

Suppose that $R = \{(Q_i, r_i) | Q_i \in C(G)\}$. In H , we assign a label $L(v) = (t(v), k(v))$ to each vertex $v \in V(H)$, where $t(v) \in Y \cup \{F\}$ with $Y = \{0, 1\}$, $L(x_i) = (t(x_i) = 0, k(x_i) = r_i)$ for every $x_i \in X$ and $L(v) = (t(v) = F, k(v) = 0)$ for every $v \in V$.

Lemma 4. [18] *Let $G = (V, E)$ be a strongly chordal graph with $|V| = n$ and $|E| = m$.*

- (1) *The vertex-clique incidence graph H of G has $O(n + m)$ edges and $O(n)$ vertices.*
- (2) *The vertex-clique incidence graph H is a strongly chordal graph and a strong elimination ordering of H can be obtained from a given one for G in $O(n + m)$ time.*

Lemma 5. *Let $G = (V, E)$ be a strongly chordal graph with $R = \{(Q_i, r_i) | Q_i \in C(G)\}$ and let H be the vertex-clique incidence graph of G with a labelling function L as mentioned above. The R -clique-transversal number of G is equal to the L -domination number of the graph H .*

Proof. Suppose that S is a minimum R -clique-transversal set of G . Let $h : V(H) \rightarrow \{0, 1\}$ be a function defined by $h(x_i) = 0$ for $x_i \in X$, $h(v) = 1$ for $v \in S$, and $h(v) = 0$ for $v \in V \setminus S$. Since $|S \cap Q_i| \geq r_i$ for every maximal clique Q_i of G , therefore $h(N_H[x_i]) = h(x_i) + h(Q_i) = h(Q_i) \geq r_i$. Clearly, $h(v) \geq 0$ for $v \in V$. The function h is an L -dominating function of H . Hence $\gamma_L(H) \leq \tau_R(G)$.

Suppose that h is a minimum L -dominating function of H . Notice that if $h(v) = 1$, then $v \in V \subseteq V(H)$. Let S be a subset of V and for every vertex $v \in V$, $v \in S$ if $h(v) = 1$. Since $h(N_H[x_i]) = h(x_i) + h(Q_i) = h(Q_i) \geq r_i$ for $x_i \in X$, therefore $|S \cap Q_i| \geq r_i$ for every maximal clique Q_i of G . The set S is an R -clique-transversal set of G . Hence $\tau_R(G) \leq \gamma_L(H)$. Following the discussion above, this lemma holds. $\square$

Theorem 6. *Let G be a strongly chordal graph. With a linear-time reduction from the generalized clique-transversal problem on G to the labelled domination problem on H as mentioned above, the generalized clique-transversal problem on G can be solved by Algorithm MLD in linear-time if the strong elimination ordering is given.*

Proof. It follows from Lemmas 4–5 and Theorem 5. $\square$

Corollary 1. *Algorithm MLD can be used to solve the clique-transversal problem, the $i,1$ clique-cover problem, the maximum q -colorable subgraph problem and the neighborhood covering problem in linear-time on strongly chordal graphs.*

Proof. Let G be a strongly chordal graph with $R = \{(Q_i, r_i) | Q_i \in C(G)\}$. If we restrict $r_i = 1$ for all maximal cliques Q_i of G , the generalized clique-transversal problem becomes the *clique-transversal problem*. According to the papers [7] and [24], the *neighborhood covering problem* is equivalent to the clique-transversal problem for strongly chordal graphs. In [8], Chang et al. showed that the generalized clique-transversal problem includes the clique-transversal problem, the $i,1$ clique-cover problem, and for perfect graphs, the maximum q -colorable subgraph problem as special cases. By Theorem 6, the clique-transversal problem, the $i,1$ clique-cover problem, the maximum q -colorable subgraph problem and the neighborhood covering problem can be solved by Algorithm MLD in linear-time on strongly chordal graphs. $\square$

3.2 NP-complete results

In this section, we present a variety of NP-complete results on doubly chordal graphs, planar graphs, and chordal bipartite graphs. Before presenting the NP-complete results, we restate the k -tuple domination, signed domination, and minus domination problems as decision problems.

(1) **The k -tuple domination problem:**

Instance: A graph $G = (V, E)$ and positive integers k and K .

Question: Is $\gamma_{\times k}(G) \leq K$?

(2) **The signed domination problem:**

Instance: A graph $G = (V, E)$ and a positive integer K .

Question: Is $\gamma_s(G) \leq K$?

(3) **The minus domination problem:**

Instance: A graph $G = (V, E)$ and a positive integer K .

Question: Is $\gamma^-(G) \leq K$?

3.2.1 Doubly chordal graphs

A graph G is *doubly chordal* if and only if G is chordal and dually chordal [27]. It is known that for a fixed positive integer $k > 1$, the k -tuple domination problem is NP-complete on split graphs and hence on chordal graphs [26].

Theorem 7. *For a fixed positive integer $k > 1$, the k -tuple domination problem is NP-complete on doubly chordal graphs.*

Proof. The k -tuple domination problem on doubly chordal graphs is clearly in NP. In the following we show the NP-completeness of the k -tuple domination problem on doubly chordal graphs by reducing the $(k-1)$ -tuple domination problem on chordal graphs to it in polynomial-time.

Let $G = (V, E)$ be a chordal graph. We construct the graph H by adding a new vertex w adjacent to all vertices of G . The construction of H can be done in polynomial-time and H is clearly a doubly chordal graph.

Claim 1. $\gamma_{\times k}(H) = \gamma_{\times k-1}(G) + 1$.

Proof. Suppose that D is a minimum $(k-1)$ -tuple dominating set of G . The number of vertices in D is at least $k-1$. By the construction of H , the vertex w is adjacent to all vertices of G . Therefore $D \cup \{w\}$ is a k -tuple dominating set of H . Then $\gamma_{\times k}(H) \leq \gamma_{\times k-1}(G) + 1$. Conversely, let D^* be a minimum k -tuple dominating set of H and let $w' \neq w$ be a vertex in D^* . Suppose that w is not in D^* . It is clear that $D^* \setminus \{w'\} \cup \{w\}$ is still a minimum k -tuple dominating set of H , so we assume that $w \in D^*$. Let $D = D^* \setminus \{w\}$. Clearly D is a $(k-1)$ -tuple dominating set of G . Then $\gamma_{\times k-1}(G) \leq \gamma_{\times k}(H) - 1$. Following the discussion above, we have $\gamma_{\times k}(H) = \gamma_{\times k-1}(G) + 1$. $\square$

The above claim implies that for a positive integer K , $\gamma_{\times k-1}(G) \leq K$ if and only if $\gamma_{\times k}(H) \leq K + 1$. $\square$

We now consider the signed domination problem on doubly chordal graphs. First, we introduce the *signed zero-domination problem* as follows:

Definition 9. A *signed zero-dominating function* of a graph $G = (V, E)$ is a function $f : V \rightarrow \{-1, 1\}$ such that $f(N_G[v]) \geq 0$ for all vertices $v \in V$. The *signed zero-domination number* of G is $\gamma_s^0(G) = \min\{f(V) \mid f \text{ is a signed zero-dominating function of } G\}$. The signed zero-domination problem is to find a signed zero-dominating function of minimum weight for G .

We restate the signed zero-domination problem as a decision problem.

The signed zero-domination problem:

Instance: A graph $G = (V, E)$ and a positive integer K .

Question: Is $\gamma_s^0(G) \leq K$?

Theorem 8. *The signed zero-domination problem is NP-complete on chordal graphs.*

Proof. The signed zero-domination problem on chordal graphs is clearly in NP. It is known that the domination problem is NP-complete on chordal graphs [2]. In the following, we show the NP-completeness of the signed zero-domination problem on chordal graphs by reducing the domination problem on chordal graphs to it in polynomial-time.

Given a chordal graph $G = (V, E)$, we construct the graph H by adding to each vertex v of G a set of $\deg_G(v)$ paths of length three, say, $v - v_{i_1} - v_{i_2} - v_{i_3}$ for $1 \leq i \leq \deg_G(v)$. That is, $V(H) = V \cup (\bigcup_{v \in V} \{v_{i_1}, v_{i_2}, v_{i_3} \mid 1 \leq i \leq \deg_G(v)\})$ and $E(H) = E \cup (\bigcup_{v \in V} \{(v, v_{i_1}), (v_{i_1}, v_{i_2}), (v_{i_2}, v_{i_3}) \mid 1 \leq i \leq \deg_G(v)\})$. Clearly, H is a chordal graph and the construction of H can be done in polynomial-time. Let $|V| = n$ and $|E| = m$. We have $|V(H)| = n + 6m$.

Claim 2. $\gamma_s^0(H) = 2m - n + 2\gamma(G)$.

Proof. Let D be a dominating set of G of $\gamma(G)$ vertices. Let $h : V(H) \rightarrow \{-1, 1\}$ be a function of H such that for each vertex $v \in V$, $h(v) = 1$ if $v \in D$, $h(v) = -1$ if $v \in V \setminus D$, and $h(v_{i_1}) = h(v_{i_2}) = 1$ and $h(v_{i_3}) = -1$ for $1 \leq i \leq \deg_G(v)$. By the construction of H , $(N_H[v] \setminus N_G[v]) = \{v_{i_1} \mid 1 \leq i \leq \deg_G(v)\}$ for every vertex $v \in V$. Since $|N_G[v] \cap D| \geq 1$ for every vertex $v \in V$, therefore h assigns the value 1 to at least one vertex in $N_G[v]$. Thus, the function h assigns the value 1 to at least $\deg_G(v) + 1$ vertices in $N_H[v]$ for every vertex $v \in V$. Since $|N_H[v]| = 2 \cdot \deg_G(v) + 1$ for every vertex $v \in V$, $h(N_H[v]) \geq 1$. For each vertex $v \in V$, it is easy to verify that $h(N_H[v_{i_1}]) \geq 0$, $h(N_H[v_{i_2}]) \geq 0$, and

$h(N_H[v_{i_3}]) \geq 0$ for $1 \leq i \leq \deg_G(v)$. Hence h is a signed zero-dominating function of H . We have $\gamma_s^0(H) \leq (\sum_{v \in V} \deg_G(v)) + (\gamma(G) - (n - \gamma(G))) = 2m - n + 2\gamma(G)$.

Conversely, let g be a minimum signed zero-dominating function of H . For each $v \in V$, we now consider a corresponding path $v - v_{i_1} - v_{i_2} - v_{i_3}$ in H , where $1 \leq i \leq \deg_G(v)$. Suppose that g assigns the value -1 to at least two vertices of v_{i_1} , v_{i_2} , and v_{i_3} . Then $g(N_H[v_{i_2}]) < 0$, which contradicts that g is a signed zero-dominating function of H . Therefore, g assigns the value -1 to at most one vertex of v_{i_1} , v_{i_2} , and v_{i_3} . Suppose that $g(v_{i_1}) = g(v_{i_2}) = g(v_{i_3}) = 1$. Then, we can obtain another signed zero-dominating function f by setting $f(v_{i_3}) = -1$ and $f(u) = g(u)$ for all remaining vertices u in $V(H)$. It contradicts the assumption that g is a minimum signed zero-dominating function of H . Hence, g assigns the value -1 to exactly one vertex of v_{i_1} , v_{i_2} , and v_{i_3} . Suppose that $g(v_{i_3}) \neq -1$. We can obtain another minimum signed zero-dominating function h of H such that $h(v_{i_1}) = h(v_{i_2}) = 1$, $h(v_{i_3}) = -1$, and $h(u) = g(u)$ for all remaining vertices u in $V(H)$. Hence, there is a minimum signed zero-dominating function h of H such that for each vertex $v \in V$, $h(v_{i_1}) = h(v_{i_2}) = 1$ and $h(v_{i_3}) = -1$ for $1 \leq i \leq \deg_G(v)$. Notice that $|N_G[v]| = \deg_G(v) + 1$ and $|N_H[v]| = 2 \cdot \deg_G(v) + 1$ for every vertex $v \in V$. The function h assigns the value 1 to at least one vertex of $N_G[v]$ for every $v \in V$. Hence, the set $\{v \in V \mid g(v) = 1\}$ is a dominating set of G . Let $K' = |\{v \in V \mid g(v) = 1\}|$. The weight of h is $\gamma_s^0(H) = (\sum_{v \in V} \deg_G(v)) + (K' - (n - K')) = 2m - n + 2K'$. We have $\gamma(G) \leq K' = \frac{\gamma_s^0(H) - (2m - n)}{2}$. Following the discussion above, $\gamma_s^0(H) = 2m - n + 2\gamma(G)$. $\square$

The above claim implies that for a positive integer K , $\gamma(G) \leq K$ if and only if $\gamma_s^0(H) \leq 2m - n + 2K$. $\square$

Theorem 9. *The signed domination problem is NP-complete on doubly chordal graphs.*

Proof. The signed domination problem on doubly chordal graphs is clearly in NP. In the following we show the NP-completeness of the signed domination problem on doubly chordal graphs by reducing the signed zero-domination problem on chordal graphs to it in polynomial-time.

Let $G = (V, E)$ be a chordal graph with $|V| = n$. We construct the graph H by adding a new vertex u adjacent to all vertices of G and adding

n new vertices $w_1, w_2, \dots, w_n$ respectively adjacent to u . The construction of H can be done in polynomial-time, and H is clearly a doubly chordal graph.

Claim 3. $\gamma_s(H) = \gamma_s^0(G) + (n + 1)$.

Proof. Suppose that g is a minimum signed zero-dominating function of G . We let $h : V(H) \rightarrow \{-1, 1\}$ be a function of H defined by $h(u) = h(w_i) = 1$ for $1 \leq i \leq n$ and $h(v) = g(v)$ if $v \in V$. Clearly, the function h is a signed dominating function of H . We have $\gamma_s(H) \leq \gamma_s^0(G) + (n + 1)$.

Suppose that h is a minimum signed dominating function of H . Then necessarily, $h(u) = h(w_i) = 1$ for $1 \leq i \leq n$. We let $g : V \rightarrow \{-1, 1\}$ be a function of G defined by $g(v) = h(v)$ if $v \in V$. By the construction of H , every vertex $v \in V$ is adjacent to u . Therefore, $g(N_G[v]) = h(N_H[v]) - 1 \geq 0$ for $v \in V$. The function g is a signed zero-dominating function of G . We have $\gamma_s^0(G) \leq \gamma_s(H) - (n + 1)$. Following the discussion above, $\gamma_s(H) = \gamma_s^0(G) + (n + 1)$. $\square$

The above claim implies that for a positive integer K , $\gamma_s^0(G) \leq K$ if and only if $\gamma_s(H) \leq K + (n + 1)$. $\square$

3.2.2 Chordal bipartite graphs and planar graphs

A *chord* is an edge between two non-consecutive vertices of a cycle. *Chordal bipartite graphs* are bipartite graphs in which every cycle of length greater than 4 has a chord. It is known that the domination problem is NP-complete on planar graphs [15] and chordal bipartite graphs [28].

Theorem 10. *The signed domination problem is NP-complete on chordal bipartite graphs and planar graphs.*

Proof. It is obvious that the signed domination problem on chordal bipartite graphs (respectively, planar graphs) is in NP. In the following we show the NP-completeness of the signed domination problem on chordal bipartite graphs (respectively, planar graphs) by reducing the domination problem on chordal bipartite graphs (respectively, planar graphs) to it in polynomial-time. The following proof is based upon a construction proposed by Hattingh et al. [20] for proving the NP-completeness of the signed domination problem on bipartite graphs.

Given a graph $G = (V, E)$, we construct the graph H by adding to each vertex v of G a set of

$\deg_G(v) + 1$ paths of length two, say, $v - v_{i_1} - v_{i_2}$ for $1 \leq i \leq \deg_G(v) + 1$. That is, $V(H) = V \cup (\bigcup_{v \in V} \{v_{i_1}, v_{i_2} \mid 1 \leq i \leq \deg_G(v) + 1\})$ and $E(H) = E \cup (\bigcup_{v \in V} \{(v, v_{i_1}), (v_{i_1}, v_{i_2}) \mid 1 \leq i \leq \deg_G(v) + 1\})$. Clearly, H is a chordal bipartite graph (respectively, planar graph) if G is a chordal bipartite graph (respectively, planar graph). Let $|V| = n$ and $|E| = m$. We have $|V(H)| = 3n + 4m$ and $|E(H)| = 2n + 5m$. Evidently, the construction of H can be done in polynomial-time.

For any signed dominating function h of H , if a vertex w is of degree 1 and $N_H(w) = \{u\}$, then $h(u) = h(w) = 1$. Therefore, $h(v) = 1$ for every vertex $v \in V(H) \setminus V$. In particular, if $h(v) = -1$, then $v \in V$. Notice that $|N_G[v]| = \deg_G(v) + 1$ and $|N_H[v]| = 2 \cdot (\deg_G(v) + 1)$ for every vertex $v \in V$. For every $v \in V$, at least one vertex of $N_G[v]$ is assigned the value 1. Hence, the set $\{v \in V \mid h(v) = 1\}$ is a dominating set of G . For a positive integer K , let $K' = |V(H)| - 2(n - K) = 4m + n + 2K$. It is easy to see that $\gamma(G) \leq K$ if and only if $\gamma_s(H) \leq K'$. $\square$

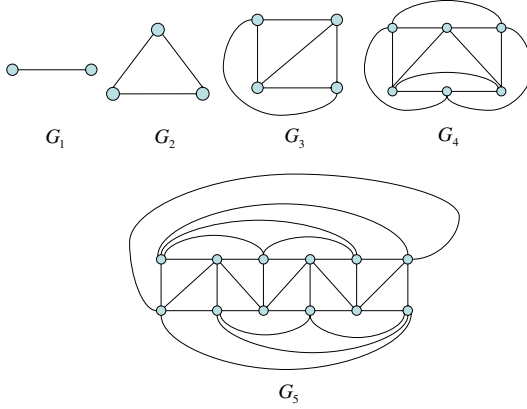
Theorem 11. *The minus domination problem is NP-complete on chordal bipartite graphs and planar graphs.*

Proof. It is obvious that the minus domination problem on chordal bipartite graphs (respectively, planar graphs) is in NP. In the following we show the NP-completeness of the minus domination problem on chordal bipartite graphs (respectively, planar graphs) by reducing the domination problem on chordal bipartite graphs (respectively, planar graphs) to it in polynomial-time. The following proof is based upon a construction proposed by Dunbar et al. [11] for proving the NP-completeness of the minus domination problem on bipartite graphs.

Given a graph $G = (V, E)$, we construct the graph H by adding a path of length 3, say $v - v_1 - v_2 - v_3$, to each vertex v of G . That is, $V(H) = V \cup (\bigcup_{v \in V} \{v_1, v_2, v_3\})$ and $E(H) = E \cup (\bigcup_{v \in V} \{(v, v_1), (v_1, v_2), (v_2, v_3)\})$. Then, H is a chordal bipartite graph (respectively, planar graph) if G is a chordal bipartite graph (respectively, planar graph). Clearly, H can be constructed in polynomial-time. Let $|V| = n$.

Claim 4. $\gamma^-(H) = \gamma(H) = \gamma(G) + n$

Proof. Among the minimum minus dominating functions of H , let h be one which assigns the value -1 to as few vertices as possible. Let $v \in V$, and let v, v_1, v_2, v_3 be the path of length 3 added to v . Clearly, $h(v_2) \geq 0$ and $h(v_3) \geq 0$. If $h(v_1) = -1$,


 Figure 1: Graphs G_1 through G_5 .

then $h(v) = h(v_2) = h(v_3) = 1$. However, the function $f : V(H) \rightarrow \{-1, 0, 1\}$ defined by $f(v_1) = f(v_2) = 0$ and $f(w) = h(w)$ for every vertex $w \in V(H) \setminus \{v_1, v_2\}$ is a minus dominating function of H of weight $h(V(H))$ that assigns the value -1 to fewer vertices than h does, contradicting the choice of h . Hence $h(v_1) \geq 0$. Furthermore, if $h(v) = -1$, then $h(v_1) = h(v_2) = 1$. However, the function $f : V(H) \rightarrow \{-1, 0, 1\}$ defined by $f(v) = f(v_1) = 0$ and $f(w) = h(w)$ for every vertex $w \in V(H) \setminus \{v, v_1\}$ is a minus dominating function of H of weight $h(V(H))$ that assigns the value -1 to fewer vertices than h does, contradicting the choice of h . Hence $h(v) \geq 0$ for every vertex $v \in V(H)$. It follows that h is a dominating function of H , so $\gamma(H) \leq h(V(H)) = \gamma^-(H)$. On the other hand, if D is a minimum dominating set of H , the function h of S (defined by $h(v) = 1$ if $v \in D$ and $h(v) = 0$ if $v \notin D$) is a minus dominating function of H , so $\gamma^-(H) \leq h(V(H)) = \gamma(H)$. Following the discussion above, $\gamma^-(H) = \gamma(H)$. It is not hard to see that $\gamma(H) = \gamma(G) + n$. This completes the proof. $\square$

The above claim implies that for a positive integer K , $\gamma(G) \leq K$ if and only if $\gamma^-(H) \leq K + n$. $\square$

Lemma 6. Let $G = (V, E)$ be a planar graph. For any positive integer $k \geq 7$, G does not have any k -tuple dominating set.

Proof. It is known that a planar graph with n vertices has at most $3n - 6$ edges. Suppose that G has a k -tuple dominating set with $k \geq 7$. Every

vertex v of G is of degree at least 6. Then, G has at least $3n$ edges, a contradiction. $\square$

Theorem 12. For a fixed positive integer k with $1 < k \leq 6$, the k -tuple domination problem is NP-complete on planar graphs.

Proof. By Lemma 6, we consider the positive integer k only for $1 < k \leq 6$. The k -tuple domination problem on planar graphs is clearly in NP. It is known that the domination problem is NP-complete on planar graphs [15]. In the following we show the NP-completeness of the k -tuple domination problem on planar graphs by reducing the $(k-1)$ -tuple domination problem on planar graphs to it in polynomial-time.

Let $G = (V, E)$ be a planar graph. We construct the graph H using the following steps:

- (1) For each vertex $v \in V$, we construct the subgraph $G_v = (V_{G_v}, E_{G_v}) = G_{k-1}$ as shown in Figure 1. Clearly, G_v is a planar graph.
- (2) Among all vertices in V_{G_v} , we pick a vertex in the outer face and call it v_1 . Let vertex v be adjacent to v_1 .

Notice that $|N_{G_v}[w]| = k$ for all vertices $w \in V_{G_v}$. Evidently, the construction of the graph H can be done in polynomial-time and H is a planar graph. Let $|V_{G_v}| = p$ and $|V| = n$.

Claim 5. $\gamma_{\times k}(H) = \gamma_{\times k-1}(G) + n \cdot p$.

Proof. Suppose that D is a minimum $(k-1)$ -tuple dominating set of G . Let $D^* = D \cup (\bigcup_{v \in V} V_{G_v})$. By the construction of H , every vertex $v \in V$ has exactly one neighbor v_1 in G_v , and $|N_H[v_1]| = k+1$ and $|N_H[w]| = k$ for all vertices $w \in V_{G_v} \setminus \{v_1\}$. Therefore, $|N_H[u] \cap D^*| \geq k$ for all vertices $u \in V(H)$. Then D^* is a k -tuple dominating set of H . We have $\gamma_{\times k}(H) \leq \gamma_{\times k-1}(G) + n \cdot p$.

Conversely, let D^* be a minimum k -tuple dominating set of H . For every vertex $v \in V$, $|N_H[v_1]| = k+1$ and $|N_H[w]| = k$ for all vertices $w \in V_{G_v} \setminus \{v_1\}$. Notice that $\bigcup_{w \in V_{G_v} \setminus \{v_1\}} N_H[w] = V_{G_v}$ for all $v \in V$. Therefore, V_{G_v} is a subset of D^* for all $v \in V$. Let $D = D^* \setminus (\bigcup_{v \in V} V_{G_v})$. Then $D \subseteq V$. By the construction of H , every vertex $v \in V$ has exactly one neighbor v_1 in G_v . Then $|N_G[v] \cap D| \geq k-1$ for every vertex $v \in V$ and D is a $(k-1)$ -tuple dominating set of G . We have $\gamma_{\times k-1}(G) \leq \gamma_{\times k}(H) - n \cdot p$. Following the discussion above, we have $\gamma_{\times k}(H) = \gamma_{\times k-1}(G) + n \cdot p$. $\square$

The above claim implies that for a positive integer K , $\gamma_{\times k-1}(G) \leq K$ if and only if $\gamma_{\times k}(H) \leq K + n \cdot p$. $\square$

Theorem 13. *For a fixed positive integer $k > 1$, the k -tuple domination problem is NP-complete on chordal bipartite graphs.*

Proof. The k -tuple domination problem on chordal bipartite graphs is clearly in NP. It is known that the domination problem is NP-complete on chordal bipartite graphs [28]. In the following we show the NP-completeness of the k -tuple domination problem on chordal bipartite graphs by reducing the $(k-1)$ -tuple domination problem on chordal bipartite graphs to it in polynomial-time.

Let $G = (V, E)$ be a chordal bipartite graph. We construct the graph H using the following steps:

- (1) For each vertex $v \in V$, we construct the subgraph $G_v = (V_{G_v}, E_{G_v})$ which is a complete bipartite graph $K_{k-1, k-1}$.
- (2) Among all vertices in V_{G_v} , we pick a vertex and call it v_1 . Let v be adjacent to v_1 .

Notice that $|N_{G_v}[w]| = k$ for all vertices $w \in V_{G_v}$. Evidently, the construction of the graph H can be done in polynomial-time and H is a chordal bipartite graph. Let $|V| = n$.

Claim 6. $\gamma_{\times k}(H) = \gamma_{\times k-1}(G) + (2k-2) \cdot n$.

Proof. Suppose that D is a minimum $(k-1)$ -tuple dominating set of G . Let $D^* = D \cup (\bigcup_{v \in V} V_{G_v})$.

By the construction of H , every vertex $v \in V$ has exactly one neighbor v_1 in G_v , and $|N_H[v_1]| = k+1$ and $|N_H[w]| = k$ for all vertices $w \in V_{G_v} \setminus \{v_1\}$. Therefore, $|N_H[u] \cap D^*| \geq k$ for all vertices $u \in V(H)$. Then D^* is a k -tuple dominating set of H . We have $\gamma_{\times k}(H) \leq \gamma_{\times k-1}(G) + (2k-2) \cdot n$.

Conversely, let D^* be a minimum k -tuple dominating set of H . For every vertex $v \in V$, $|N_H[v_1]| = k+1$ and $|N_H[w]| = k$ for all vertices $w \in V_{G_v} \setminus \{v_1\}$. Notice that $\bigcup_{w \in V_{G_v} \setminus \{v_1\}} N_H[w] = V_{G_v}$ for all $v \in V$. Therefore, V_{G_v} is a subset of D^* for all $v \in V$. Let $D = D^* \setminus (\bigcup_{v \in V} V_{G_v})$. Then

$D \subseteq V$. By the construction of H , every vertex $v \in V$ has exactly one neighbor v_1 in G_v . Then $|N_G[v] \cap D| \geq k-1$ for every vertex $v \in V$ and D is a $(k-1)$ -tuple dominating set of G . We have $\gamma_{\times k-1}(G) \leq \gamma_{\times k}(H) - (2k-2) \cdot n$. Following the discussion above, we have $\gamma_{\times k}(H) = \gamma_{\times k-1}(G) + (2k-2) \cdot n$. $\square$

The above claim implies that for a positive integer K , $\gamma_{\times k-1}(G) \leq K$ if and only if $\gamma_{\times k}(H) \leq K + (2k-2) \cdot n$. $\square$

4 Concluding remarks

The algorithm for labelled domination on strongly chordal graphs provides a unified approach to the $\{k\}$ -domination problem, the k -tuple domination problem, the signed domination problem, the minus domination problem, and the generalized clique-transversal problem on strongly chordal graphs in linear-time. Furthermore, the algorithm solves other variations of domination for strongly chordal graphs. For example, the algorithm can solve the following two generalized problems of the signed and minus domination problems for strongly chordal graphs.

Definition 10. For a fixed positive integer k , a k -signed dominating function of a graph $G = (V, E)$ is a function $f : V \rightarrow \{-1, 1\}$ such that $f(N_G[v]) \geq k$ for all vertices $v \in V$. The k -signed domination problem is to find a k -signed dominating function of minimum weight for a graph G .

Definition 11. For a fixed positive integer k , a k -minus dominating function of a graph $G = (V, E)$ is a function $f : V \rightarrow \{-1, 0, 1\}$ such that $f(N_G[v]) \geq k$ for all vertices $v \in V$. The k -minus domination problem is to find a k -minus dominating function of minimum weight for a graph G .

We close the paper with the following two questions. What is the complexity of the minus domination problem on doubly chordal graphs? What is the complexity of the $\{k\}$ -domination problem when restricted to chordal bipartite graphs and planar graphs?

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References

- [1] D.W. Bange, A.E. Barkauskas, L.H. Host, and P.J. Slater, Generalized domination and efficient domination in graphs, *Discrete Math.*, **159** (1996) 1–11.

- [2] A.A. Bertossi, Dominating sets for split and bipartite graphs, *Inform. Process. Lett.*, **19**(1) (1984) 37–40.
- [3] A. Brandstädt, V.D. Chepoi, F.F. Dragan, The algorithmic use of hypertree structure and maximum neighbourhood orderings, *Discrete Appl. Math.*, **82** (1998) 43–77.
- [4] A. Brandstädt, F.F. Dragan, V.D. Chepoi, and V. Voloshin, Dually chordal graphs, *SIAM J. Discrete Math.*, **11** (1998) 437–455.
- [5] A. Brandstädt, V.B. Le, and J.P. Spinrad, Graph Classes: A Survey, SIAM Monographs on Discrete Mathematics and Applications, Philadelphia 1999.
- [6] B. Brešar, M.A. Henning, and S. Klavžar, On integer domination in graphs and vizing-like problems, To appear in Taiwanese J. Math., 2006.
- [7] G.J. Chang, M. Farber, and Zs. Tuza, Algorithmic aspects of neighborhood numbers, *SIAM J. Discrete Math.*, **6**(1) 24–29.
- [8] M.S. Chang, Y.H. Chen, G.J. Chang, and J.H. Yan, Algorithmic aspects of the generalised clique transversal problem on chordal graphs, *Discrete Appl. Math.* **66** (1996) 189–203.
- [9] G.J. Chang, Algorithmic aspects of domination in graphs, in: D.Z. Du, P.M. Pardalos (Eds.), Handbook of Combinatorial Optimization, Vol. 3, Kluwer, Boston, MA, 1998, pp. 339–405.
- [10] Y.M. Chen, The fault tolerant domination problem on strongly chordal graphs, Master Thesis, National Chung Cheng University, Taiwan, July 2001.
- [11] J. Dunbar, W. Goddard, S. Hedetniemi, A. McRae, and M.A. Henning, The algorithmic complexity of minus domination in graphs, *Discrete Appl. Math.*, **68** (1996) 73–84.
- [12] J. Dunbar, S. Hedetniemi, M.A. Henning, and A. McRae, Minus domination in graphs, *Discrete Math.*, **199** (1999) 35–47.
- [13] F.F. Dragan, HT-graphs: centers, connected r -domination, and Steiner trees, *Comput. Sci. J. Moldova*, **1** (1993) 64–83.
- [14] M. Farber, Characterizations of strongly chordal graphs, *Discrete Math.* **43** (1983) 173–189.
- [15] M.R. Garey and D.S. Johnson, Computers and Intractability : A Guide to the Theory of NP-Completeness, W. H. Freeman, San Francisco, CA, 1979.
- [16] M. Gairing, S.T. Hedetniemi, P. Kristiansen, and A.A. McRae, Self-stabilizing algorithms for $\{k\}$ -domination, in: Proceedings of the 6th Symposium on Self-Stabilization (SSS2003), Lecture Notes in Computer Science 2704, pp. 49–60, 2003.
- [17] H.J. Gong, Signed domination number in block graphs, Master Thesis, National Central University, Taiwan, June 2004.
- [18] V. Guruswami, and C.P. Rangan, Algorithmic aspects of clique-transversal and clique-independent sets, *Discrete Appl. Math.* **100** (2000) 183–202.
- [19] F. Harary, T.W. Haynes, Double domination in graphs, *Ars Combin.*, **55** (2000) 201–213.
- [20] J.H. Hattingh, M.A. Henning, and P.J. Slater, The algorithmic complexity of signed domination in graphs, *Australas. J. Combin.*, **12** (1995) 101–112.
- [21] T.W. Haynes, S.T. Hedetniemi, and P.J. Slater, Fundamentals of Domination in Graphs, Marcel Dekker, New York, 1998.
- [22] T.W. Haynes, S.T. Hedetniemi, and P.J. Slater, Domination in Graphs: Advanced Topics, Marcel Dekker, New York, 1998.
- [23] M.A. Henning, Dominating functions in graphs, Chapter 2 in: Domination in Graphs: Advanced Topics, T. Haynes, S. Hedetniemi, P. Slater, (eds.), Marcel Dekker, 1998, 31–89.
- [24] J. Lehel and Zs. Tuza, Neighborhood perfect graphs, *Discrete Math.*, **61** (1986) 93–101.
- [25] C.S. Liao and G.J. Chang, Algorithmic aspect of k -tuple domination in graphs, *Taiwanese J. Math.*, **6**(3) (2002) 415–420.
- [26] C.S. Liao and G.J. Chang, k -tuple domination in graphs, *Inform. Process. Lett.*, **87** (2003) 45–50.

- [27] M. Moscarini, Doubly chordal graphs, Steiner trees, and connected domination, *Networks*, **23** (1993) 59–69.
- [28] H. Müller and A. Brandsädt, The NP-Completeness of Steiner tree and dominating set for chordal bipartite graphs, *Theoret. Comput. Sci.*, **53** (1987) 257–265.
- [29] R. Paige and R.E. Tarjan, Three partition refinement algorithms, *SIAM J. Comput.*, **16** (1987) 973–989.
- [30] D.J. Rose, Triangulated graphs and the elimination process, *J. Math. Anal. Appl.*, **32** (1970) 597–609.
- [31] J.P. Spinrad, Doubly lexical ordering of dense 0-1 matrices, *Inform. Process. Lett.*, **45** (1993) 229–235.
- [32] H.G. Yeh and G.J. Chang, Algorithmic aspects of majority domination, *Taiwanese J. Math.*, **1** (3) (1997), 343–350.